Frobenius-equivariant pair extension of eight-arcs in $\mathrm{PG}(2, 25)$

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Abstract

Let ϕ be the quadratic Frobenius involution of $\mathrm{PG}(2, s^2)$. We give an exact carrier-wise count for fresh conjugate pairs $\{P, \phi(P)\}$ that extend a ϕ-invariant arc. The count combines the exact number of empty Baer lines with a charge from noninvariant secant orbits. An exact correction separates secant orbits whose fixed centers make them invisible on a carrier from genuine collisions between visible charges, and it classifies equality and every first-order excess level.

We then prove that every ϕ-invariant eight-arc in $\mathrm{PG}(2, 25)$ admits a fresh conjugate-pair extension. The zero- and four-fixed-point profiles follow from incidence and moment arguments. The exceptional two-fixed-point profile is reduced, by projective normalization, to a finite certificate checked by the Lean kernel. The full theorem, the semantic global pair count, and the profile split are formalized in Lean without unproved declarations, custom axioms, or native-code evaluation. Together with the general count, the order-five result covers every prime-power base-field order $s \geq 5$.

1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. Ordinary completeness asks whether one further point can be adjoined. If the arc is preserved by a field automorphism, the natural equivariant continuation unit is instead a complete Galois orbit. Over a quadratic extension this means either a fixed point or a nonfixed conjugate pair.

Let $F = \mathbb{F}_s$, let $K = \mathbb{F}_{s^2}$, and let

$$\phi\colon [x : y : z] \longmapsto [x^s : y^s : z^s]$$

act on $\mathrm{PG}(2, K)$. The fixed points and fixed lines form the embedded Baer subplane $\mathrm{PG}(2, F)$. This paper studies the following question: when must a ϕ-invariant arc admit an extension by a fresh pair $\{P, \phi(P)\}$?

The classical ingredients are elementary: point and line counts in a projective plane, the fixed Baer subplane, Hilbert–90 normalization, and secant counting; see Hirschfeld [8] and Martin [13]. The contribution here is their exact assembly into an orbit-valued extension count. The main results are:

(i) an explicit lower bound for the number of legal conjugate-pair extensions of an arbitrary invariant arc;

(ii) an exact linewise correction that separates invisible secant centers from charge collisions, together with equality and excess criteria;

(iii) a uniform theorem that every invariant eight-arc in $\mathrm{PG}(2, 25)$ extends by a fresh conjugate pair and, together with the general estimate, covers every prime-power base order $s \geq 5$; and

(iv) a square-root lower bound for arcs that are saturated with respect to conjugate-pair insertion.

The last constant is not new: it is the classical Lunelli–Sce $\sqrt{2q}$ scale [11, 5], now obtained under the weaker hypothesis of no equivariant pair extension. We make no sharpness claim.

Prior art and claim boundary

Baker and Wantz [4] use Frobenius-invariant arcs in the Hughes plane and, in a maximality argument, consider a point together with its Frobenius image. This is genuine precedence for the conjugate-point maneuver. Their result does not count legal conjugate pairs on empty Baer lines. Jungnickel's arcs in Baer-semiplane complements [9] and Ueberberg's geometries of Frobenius collineations [16] are also adjacent but concern different incidence or collineation problems.

Ordinary arc and MDS lengthening results [1, 3, 2] impose no quadratic Galois-pair constraint. Likewise, the classification of Galois-invariant subsets of $\mathbf{P}^1$ [10] does not treat arbitrary plane arcs or their pair extensions. Bounded zbMATH Open, OpenAlex, Crossref, and source-level searches found no exact precursor for the quantitative criterion below or for the uniform $\mathrm{PG}(2,25)$ theorem. This is negative search evidence, not a claim of historical priority.

2 Quadratic Frobenius and mate lines

Let $\mathcal{C} \subseteq \mathrm{PG}(2,K)$ be a ϕ-invariant k-arc. Write

$$f = |\mathcal{C} \cap \mathrm{PG}(2,F)|, \qquad k = f + 2e,$$

so that $\mathcal{C}$ is the disjoint union of f fixed points and e nonfixed conjugate pairs.

For every nonfixed point P, the line $P\phi(P)$ is fixed by ϕ; we call it the *mate line* of the pair $\{P, \phi(P)\}$. Conversely, every nonfixed conjugate pair has a unique mate line. A fixed line contains s^2+1 points, of which $s+1$ are fixed, and therefore contains exactly

$$N = \frac{s^2 - s}{2}$$

nonfixed conjugate pairs.

Definition 2.1. A *legal conjugate-pair extension* of $\mathcal{C}$ is an unordered pair $Q = \{P, \phi(P)\}$ such that $P \neq \phi(P)$, $Q \cap \mathcal{C} = \varnothing$, and $\mathcal{C} \cup Q$ is an arc. Let $N_{\mathrm{pair}}(\mathcal{C})$ be the number of such pairs.

Lemma 2.2 (Pair-extension test)**.** *Let $P \neq \phi(P)$ and suppose neither point belongs to $\mathcal{C}$. Then $\mathcal{C} \cup \{P, \phi(P)\}$ fails to be an arc if and only if at least one of the following holds:*

(a) P lies on a secant of $\mathcal{C}$;

(b) $\phi(P)$ lies on a secant of $\mathcal{C}$;

(c) the mate line $P\phi(P)$ contains a point of $\mathcal{C}$.

Proof. Every collinear triple in the enlarged set either contains one new point and two old points, giving (a) or (b), or contains both new points and one old point, giving (c). The converse is immediate. $\square$

The useful carriers are therefore the fixed lines disjoint from $\mathcal{C}$. Let $\mathcal{E}$ be this set of empty fixed lines.

Lemma 2.3 (Empty fixed lines)**.** *The number of empty fixed lines is*

$$E = |\mathcal{E}| = s^2 + s + 1 - \left(f(s+1) - \binom{f}{2} + e \right). \tag{1}$$

Proof. The f fixed selected points occupy $f(s+1) - \binom{f}{2}$ fixed lines. Inclusion–exclusion is exact because no three selected fixed points are collinear. Each selected conjugate pair occupies its mate line. These e lines are distinct and contain no fixed selected point, since $\mathcal{C}$ is an arc. Subtract from the $s^2 + s + 1$ fixed lines. □

Among the $\binom{k}{2}$ old secants, exactly

$$I = \binom{f}{2} + e$$

are fixed: those joining two fixed selected points and the e mate lines. The remaining secants occur in two-element Frobenius orbits. Their number is

$$M = \frac{\binom{k}{2} - I}{2} = fe + e(e-1). \tag{2}$$

3 The quantitative pair-extension criterion

Theorem 3.1 (Quadratic pair-extension criterion)**.** *Let $\mathcal{C}$ be a ϕ-invariant k-arc in* $\mathrm{PG}(2, s^2)$ *with profile $k = f + 2e$. With E, N, and M as in* (1) *and* (2)*,*

$$N_{\text{pair}}(\mathcal{C}) \geq E\ (N - M)_+ = E \left(\frac{s^2 - s}{2} - fe - e(e-1) \right)_+ . \tag{3}$$

In particular, if $E > 0$ and $M < N$, then $\mathcal{C}$ admits a legal conjugate-pair extension.

Proof. Fix $\ell \in \mathcal{E}$. It carries exactly N candidate conjugate pairs. A fixed old secant meets ℓ in a fixed point and destroys no nonfixed candidate. Consider a nonfixed old secant orbit $\{m, \phi(m)\}$. Unless $m \cap \ell$ is fixed, the two intersections $m \cap \ell$ and $\phi(m) \cap \ell$ form one conjugate candidate pair; thus the orbit destroys at most one candidate. There are M such secant orbits, so at least $(N - M)_+$ candidates survive on ℓ.

A survivor lies on no old secant, and its empty mate line contains no point of $\mathcal{C}$. Lemma 2.2 makes it legal. Candidate sets on distinct fixed lines are disjoint because a nonfixed pair has a unique mate line. Summing over the E carriers proves (3). □

The preceding proof deliberately uses only the uniform upper bound M on the number of forbidden candidates per carrier. Retaining the actual forbidden support gives both a sharper bound and an exact identity.

For $\ell \in \mathcal{E}$, let $\mathcal{Q}_\ell$ be its N candidate pairs and let $\mathcal{F}_\ell \subseteq \mathcal{Q}_\ell$ be the pairs hit by at least one nonfixed old secant orbit. Then

$$N_{\text{pair}}(\mathcal{C}) = \sum_{\ell \in \mathcal{E}} \left(|\mathcal{Q}_\ell| - |\mathcal{F}_\ell| \right). \tag{4}$$

The same formula applies when carrier capacities vary. In particular, if $|\mathcal{F}_\ell| \leq U_\ell$, then

$$N_{\text{pair}}(\mathcal{C}) \geq \sum_{\ell \in \mathcal{E}} \left(|\mathcal{Q}_\ell| - U_\ell \right)_+ .$$

4 Invisible centers and collision correction

Fix $\ell \in \mathcal{E}$. Every nonfixed secant orbit has a fixed center $m \cap \phi(m)$. Let B_ℓ be the number of such orbits whose center lies on ℓ. Those orbits meet ℓ only in a fixed point and are invisible to its nonfixed candidates.

For $Q \in \mathcal{Q}_\ell$, let $\mu_\ell(Q)$ be the number of visible secant orbits that charge Q, and put

$$A_\ell = \sum_{Q \in \mathcal{Q}_\ell} \mu_\ell(Q), \qquad R_\ell = \sum_{Q \in \mathcal{Q}_\ell} (\mu_\ell(Q) - 1)_+ \,.$$

Here R_ℓ is the redundancy caused by several visible obstruction orbits hitting the same candidate.

Proposition 4.1 (Exact linewise correction)**.** *For every empty fixed line ℓ,*

$$A_\ell = M - B_\ell, \tag{5}$$
$$|\mathcal{F}_\ell| = A_\ell - R_\ell, \tag{6}$$
$$|\mathcal{Q}_\ell \setminus \mathcal{F}_\ell| + M = N + B_\ell + R_\ell. \tag{7}$$

Consequently the first-order identity $|\mathcal{Q}_\ell \setminus \mathcal{F}_\ell| + M = N$ holds exactly when every secant orbit is visible on ℓ and the visible charge map is injective.

Proof. Exactly B_ℓ of the M secant orbits are invisible, proving (5). The support of the multiplicity function μ_ℓ is $\mathcal{F}_\ell$, and for every positive integer a one has $1 = a - (a-1)$. Summing this equality over the support gives (6). Substitution into $|\mathcal{Q}_\ell \setminus \mathcal{F}_\ell| = N - |\mathcal{F}_\ell|$ gives (7). Equality in the first-order identity is equivalent to $B_\ell = R_\ell = 0$. ☐

Let

$$L = N_{\mathrm{pair}}(\mathcal{C}), \qquad B = \sum_{\ell \in \mathcal{E}} B_\ell, \qquad R = \sum_{\ell \in \mathcal{E}} R_\ell.$$

Corollary 4.2 (Aggregate equality and excess)**.** *The exact aggregate balance is*

$$L + EM = EN + B + R. \tag{8}$$

Thus $L + EM = EN$ if and only if every secant-orbit center avoids every empty fixed carrier and every local visible charge is injective. More generally, for each integer $t \geq 0$,

$$L + EM = EN + t \quad \Longleftrightarrow \quad B + R = t.$$

This is an algebraic equality and excess classification. It does not classify the invariant arcs for which L is small, and no such structural inverse theorem is claimed.

5 Eight-arcs over $\mathbb{F}_{25}$

We now take $s = 5$, so each empty fixed line carries $N = 10$ candidate pairs. The profile of an invariant eight-arc is

$$(f, e) \in \{(0,4), (2,3), (4,2), (6,1), (8,0)\}.$$

Table 1 records the numerical data and the proved lower bound for each profile. In the exceptional row the bound $L \geq 1$ records the kernel-checked existence theorem; the stronger observed census minimum in Remark 5.4 is not used.

The following simple center estimate is useful. A secant orbit joining two distinct selected conjugate pairs has a fixed center that lies on neither participating mate line. At most f

f	e	E	M	proved $L \geq$	method
0	4	27	12	5	secant-index moments
2	3	17	12	1	kernel-checked certificate
4	2	11	10	4	fixed-center incidence
6	1	9	6	36	uniform criterion
8	0	11	0	110	uniform criterion

Table 1: The five profiles of invariant eight-arcs in $\mathrm{PG}(2,25)$. Here $L = N_{\text{pair}}(\mathcal{C})$.

fixed-point star lines and $e-2$ other mate lines through this center are occupied. Since a fixed point lies on $s+1$ fixed lines, each such cross-pair secant orbit is invisible on at least

$$s + 3 - f - e \tag{9}$$

empty carriers.

Proposition 5.1 (The four nonexceptional profiles)**.** *Let $\mathcal{C}$ be an invariant eight-arc in* $\mathrm{PG}(2,25)$.

(a) If $f=8$ or $f=6$, then $L \geq 110$ or $L \geq 36$, respectively.

(b) If $f=4$, then $L \geq 4$.

(c) If $f=0$, then $L \geq 5$.

Proof. For $(f,e)=(8,0)$ one has $(E,M)=(11,0)$, and for $(6,1)$ one has $(E,M)=(9,6)$. Theorem 3.1 gives 110 and 36.

For $(f,e)=(4,2)$ one has $(E,M,N)=(11,10,10)$. There are two cross-pair secant orbits, each invisible on at least two empty carriers by (9). Hence $B \geq 4$, and (8) gives $L = B + R \geq 4$.

It remains to treat $(f,e)=(0,4)$. Here

$$(E,M,N) = (27,12,10).$$

All twelve nonfixed secant orbits are cross-pair orbits, and (9) gives $B \geq 48$.

We recall the second secant-index moment for an eight-arc:

$$\sum_{x \notin \mathcal{C}} \binom{r_x}{2} = 3\binom{8}{4} = 210, \tag{10}$$

where r_x is the number of secants of $\mathcal{C}$ through x. At fixed external points, the first secant-index sum is 48: the four fixed mate secants contain six fixed points each, contributing 24 incidences, and the other 24 secants each contain their unique fixed orbit center. Since $r_x \leq 4$ and $2\binom{n}{2} \leq 3n$ for $0 \leq n \leq 4$, their contribution to (10) is at most 72.

The four occupied fixed carriers are precisely the selected mate lines. For an external nonfixed point x on one of them, write $r_x = 1 + u_x$, where the first secant is the mate line. Every nonfixed secant meets at most the two selected mate lines not containing its endpoints, so $\sum u_x \leq 48$. Since $u_x \leq 3$ and $\binom{1+u}{2} \leq 2u$, the occupied-carrier contribution is at most 96. Thus the two endpoints of candidates on empty carriers contribute at least $210-72-96=42$ to (10). Conjugate endpoints contribute equally, and therefore

$$T := \sum_{\ell \in \mathcal{E}} \sum_{Q \in \mathcal{Q}_\ell} \binom{\mu_\ell(Q)}{2} \geq 21.$$

Because $\mu_\ell(Q) \leq 4$ and $\binom{n}{2} \leq 2(n-1)$ for $1 \leq n \leq 4$, we have $T \leq 2R$, whence $R \geq 11$. Finally, (8) gives

$$L = 27(10 - 12) + B + R \geq -54 + 48 + 11 = 5.$$

□

The remaining profile $(f, e) = (2, 3)$ has $(E, M, N) = (17, 12, 10)$. The center estimate gives $B \geq 18$, but the aggregate identity would still require a further collision estimate. We close this profile by a finite reduction.

Proposition 5.2 (Exceptional profile)**.** *Every invariant eight-arc in* $\mathrm{PG}(2, 25)$ *with exactly two fixed selected points has a fresh legal conjugate-pair extension.*

Computer-assisted proof, kernel checked. Represent $\mathbb{F}_{25}$ as $\mathbb{F}_5[\omega]/(\omega^2 - 2)$. A base-field projectivity sends the two fixed selected points to a prescribed ordered pair. Their stabilizer then sends the first admissible selected conjugate pair to

$$[1 : \omega : \omega], \qquad [1 : -\omega : -\omega].$$

Both transformations commute with Frobenius and preserve the arc and pair-extension predicates.

After these reductions, the remaining finite slice has 46,056 rows. Of these, 39,012 carry a checked witness that the proposed old set is not an arc, while 7,044 carry a checked fresh legal-pair witness. The rows are split into 1,639 leaf modules and 303 aggregate modules. Each leaf is evaluated by Lean's kernel reduction. The projective transport theorem maps the checked survivor back to the original arc and states explicitly that both new points are fresh and that the enlarged set is an arc. Section 7 records the trust boundary. □

Theorem 5.3 (Uniform order-five extension)**.** *Every Frobenius-invariant eight-arc in* $\mathrm{PG}(2, 25)$ *admits an extension by a fresh nonfixed point together with its Frobenius conjugate.*

Proof. The fixed-point count is even and at most eight. Proposition 5.1 treats $f = 0, 4, 6, 8$, and Proposition 5.2 treats $f = 2$. □

The complete-arc classifications in [12, 6] show that the smallest complete arcs in $\mathrm{PG}(2, 25)$ have size 12. Thus every eight-arc has an ordinary one-point extension. That fact does not imply Theorem 5.3: ordinary extendability neither makes the legal-point set Frobenius-stable nor ensures that a legal point and its conjugate are jointly legal.

Remark 5.4 (External census)*.* An external enumeration after fixed-point normalization reports 469,600 invariant eight-arcs with profile $(f, e) = (2, 3)$ and observed minimum legal pair count 32. Independent C++ and Python implementations agree on the census, minimum, minimizing orbit indices, and coordinate witness. These numbers are computational data, not inputs to Proposition 5.2 or Theorem 5.3.

6 Equivariant saturation

Call an invariant arc *pair-saturated* if it has no legal conjugate-pair extension. This is weaker than ordinary completeness.

Corollary 6.1 (Orbit-saturation bound)**.** *If a* ϕ*-invariant* k*-arc in* $\mathrm{PG}(2, s^2)$ *is pair-saturated, then*

$$k \geq 1 + \left\lceil \sqrt{2s(s-1)} \right\rceil. \tag{11}$$

In particular, for every prime power $s \geq 7$, every invariant eight-arc in $\mathrm{PG}(2, s^2)$ *admits a conjugate-pair extension.*

Proof. If every fixed line is occupied, Lemma 2.3 gives $s^2 + s + 1 = f(s+1) - \binom{f}{2} + e$. Substituting $2e = k - f$ and completing the square gives the occupation identity

$$k = s^2 + 1 + (f - s - 1)^2,$$

which is stronger than (11). Otherwise choose an empty fixed line. Pair saturation forces all $N = s(s-1)/2$ of its candidates to be forbidden, so $N \leq M$. From $k = f + 2e$ and (2),

$$M = e((k-1) - e), \qquad 4M \leq (k-1)^2.$$

Consequently $2s(s-1) \leq (k-1)^2$, which gives (11).

For $k = 8$, one has $M \leq 12 < N$ when $s \geq 7$. Full occupation is impossible because $8 < s^2 + 1$, so Theorem 3.1 applies. ☐

Corollary 6.2 (All base orders from five)**.** *Let $s \geq 5$ be a prime power. Every Frobenius-invariant eight-arc in* $\mathrm{PG}(2, s^2)$ *admits an extension by a fresh nonfixed point together with its Frobenius conjugate.*

Proof. The case $s = 5$ is Theorem 5.3. There is no prime power strictly between 5 and 7, and the final assertion of Corollary 6.1 treats every $s \geq 7$. ☐

The scale in (11) is the classical Lunelli–Sce scale for ordinary complete arcs, with the same secant-covering mechanism in the Frobenius quotient. Ng and Wild's line-covering problem [14] is another adjacent square-root-scale result. The point of Corollary 6.1 is the weaker pair-saturation hypothesis, not a new constant or a sharp family.

7 Formal verification

The results were formalized in Lean [7] on top of mathlib [15]. The supplementary source is divided as follows.

1. The projective-conjugation and quadratic-Frobenius modules construct the semilinear projective involution, prove the degree-two fixed-locus normalization, and count the $N = (s^2 - s)/2$ candidates on a fixed line.

2. The line-counting and forbidden-candidate modules prove Lemma 2.3, the exact secant-orbit count, the injective charge, and the semantic arc-extension conclusion in Theorem 3.1.

3. The global-count module defines the finite set of fresh nonfixed Frobenius pairs whose union with the old arc remains an arc. It proves that this semantic set is the disjoint union of the carrierwise legal sets and that its cardinality is the abstract legal count.

4. The collision and invisibility modules prove Proposition 4.1, Corollary 4.2, the fixed-center interpretation, and the cross-pair estimate (9).

5. Separate profile modules prove the lower bounds 5 and 4 for $f = 0$ and $f = 4$. The exceptional certificate proves Proposition 5.2; the aggregate module proves Theorem 5.3 by the exhaustive parity split.

6. The orbit-saturation module proves the denominator-free inequality $2s(s-1) \le (k-1)^2$ used in Corollary 6.1. The final ceiling notation and the geometric case split are paper proofs.

The scoped builds and declaration-level axiom audits report exactly

$$[\texttt{propext}, \texttt{Classical.choice}, \texttt{Quot.sound}],$$

the standard classical quotient profile inherited from mathlib. No public theorem depends on an admitted proof, a custom axiom, or `native_decide`. The finite leaves in Proposition 5.2 use kernel `decide`; the external census in Remark 5.4 is outside the theorem dependency graph.

For reproducibility, the paper-facing declarations are located as follows.

Source module	Paper-facing declaration or role
`FiniteGeom/BaerCompletion/PairExtension.lean`	`quadraticBaer_pairExtension_lowerBound`
`RelativeConicArcs/QuadraticForbidden.lean`	`exists_quadratic_pair_extension`
`RelativeConicArcs/QuadraticGlobalCount.lean`	`card_globalLegalPairs_eq_legalCount`
`RelativeConicArcs/QuadraticInvisible.lean`	aggregate equality/excess and fixed-center criteria
`RelativeConicArcs/Q25ProfileZero.lean`	`five_le_sum_card_legal_profile_zero`
`RelativeConicArcs/Q25ProfileFour.lean`	`four_le_sum_card_legal_profile_four`
`RelativeConicArcs/Q25PairResult.lean`	`f2_pair_extension`
`RelativeConicArcs/Q25AllProfiles.lean`	`pair_extension`
`FiniteGeom/BaerCompletion/OrbitSaturation.lean`	`orbitSaturation_quadratic_bound_of_split`

Artifact and reproduction

The accompanying source archive pins Lean to `leanprover/lean4:v4.32.0-rc1` and mathlib to revision `571b8a8e54219b4d393f75f4b8653fac08197fcc`. The audited formal-source checkpoint is Git revision `8f308c3d94652c3d3a66335701f1a18b53740675`. From the `lean/` directory the two paper-facing aggregate targets are rebuilt by

```
LEAN_NUM_THREADS=6 nix develop --command lake build \
  RelativeConicArcs.QuadraticGlobalCount \
  RelativeConicArcs.Q25AllProfiles
```

The manuscript PDF is reproduced from its source directory with

```
nix shell nixpkgs#tectonic -c tectonic \
  frobenius_pair_extension.tex --keep-logs
```

8 Conclusion

The fixed mate line of a quadratic Frobenius orbit turns symmetry-preserving arc extension into a carrierwise counting problem. The uniform count is a first-order obstruction bound;

its exact correction records which secant orbits disappear at fixed centers and which visible obstructions collide. This suffices, together with one finite exceptional-profile certificate, to prove a uniform conjugate-pair extension theorem for invariant eight-arcs in $\mathrm{PG}(2,25)$.

The natural strengthening is now family-specific rather than generic: prove that every invariant eight-arc has at least two legal pairs, which would give an alternate-orbit repair theorem for invariant ten-arcs after deletion of one selected nonfixed orbit. That multiplicity statement is not used or claimed here.

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